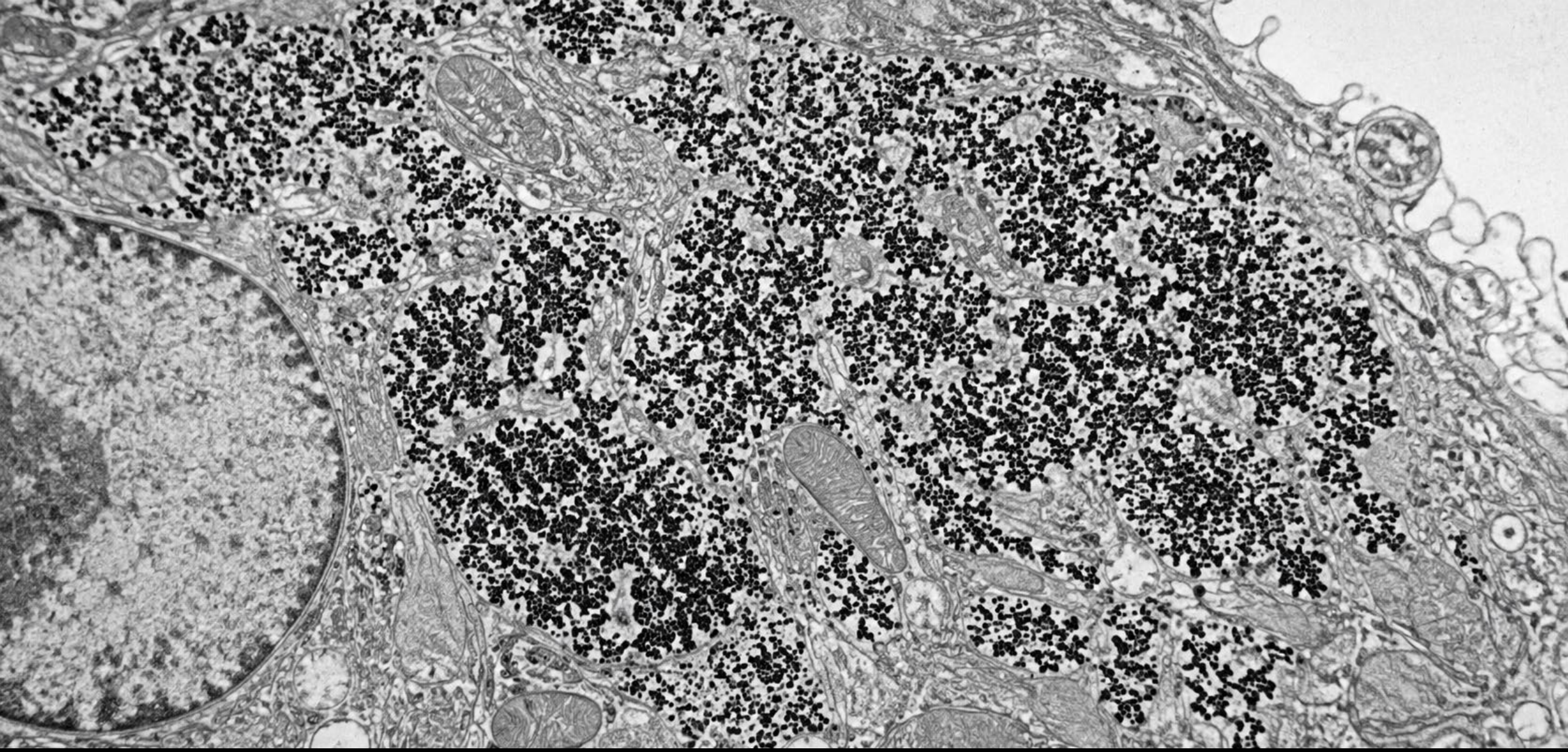
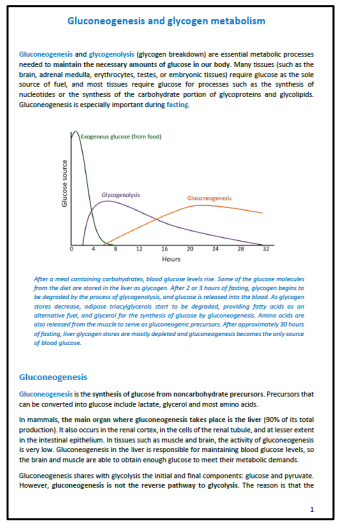


Gluconeogenesis and glycogen metabolism

Glucose homeostasis

Supportive material


Gluconeogenesis and glycogen metabolism

Gluconeogenesis and glycogenolysis (glycogen breakdown) are essential metabolic processes needed to maintain the necessary amounts of glucose in our body. Many tissues (such as the brain, adrenal medulla, erythrocytes, testes, or embryonic tissues) require glucose as the sole source of fuel, and most tissues require glucose for processes such as the synthesis of nucleotides or the synthesis of the carbohydrate portion of glycoproteins and glycolipids. Gluconeogenesis is especially important during fasting.

After a meal containing carbohydrates, blood glucose levels rise. Some of the glucose molecules from the diet are stored in the liver as glycogen. After 2 or 3 hours of fasting, glycogen begins to be degraded by the process of glycogenolysis, and glucose is released into the blood. As glycogen stores decrease, adipose triacylglycerols start to be degraded, providing fatty acids as an alternative fuel, and glycerol for the synthesis of glucose by gluconeogenesis. Amino acids are also released from the muscle to serve as gluconeogenic precursors. After approximately 30 hours of fasting, liver glycogen stores are mostly depleted and gluconeogenesis becomes the only source of blood glucose.

Gluconeogenesis

Gluconeogenesis is the synthesis of glucose from noncarbohydrate precursors. Precursors that can be converted into glucose include lactate, glycerol and most amino acids.

In mammals, the main organ where gluconeogenesis takes place is the liver (90% of its total production). It also occurs in the renal cortex, in the cells of the renal tubule, and at lesser extent in the intestinal epithelium. In tissues such as muscle and brain, the activity of gluconeogenesis is very low. Gluconeogenesis in the liver is responsible for maintaining blood glucose levels, so the brain and muscle are able to obtain enough glucose to meet their metabolic demands.

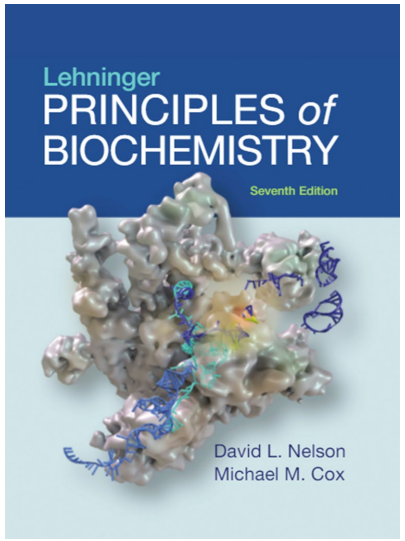
Gluconeogenesis shares with glycolysis the initial and final components: glucose and pyruvate. However, gluconeogenesis is not the reverse pathway to glycolysis. The reason is that the

1

Additional study material

ADI document:

Gluconeogenesis and glycogen metabolism



Reading

Lehninger’s PoB Chapter 14:

14.4.: Gluconeogenesis

Lehninger’s PoB Chapter 15:

15.3.: Coordinated regulation of glycolysis and gluconeogenesis

Learning goals

- Main features of gluconeogenesis
- Regulation of gluconeogenesis
- Enzymes involved in glycogen metabolism
- Hormonal regulation of glycogen metabolism
- Pathological importance of glycogen metabolism

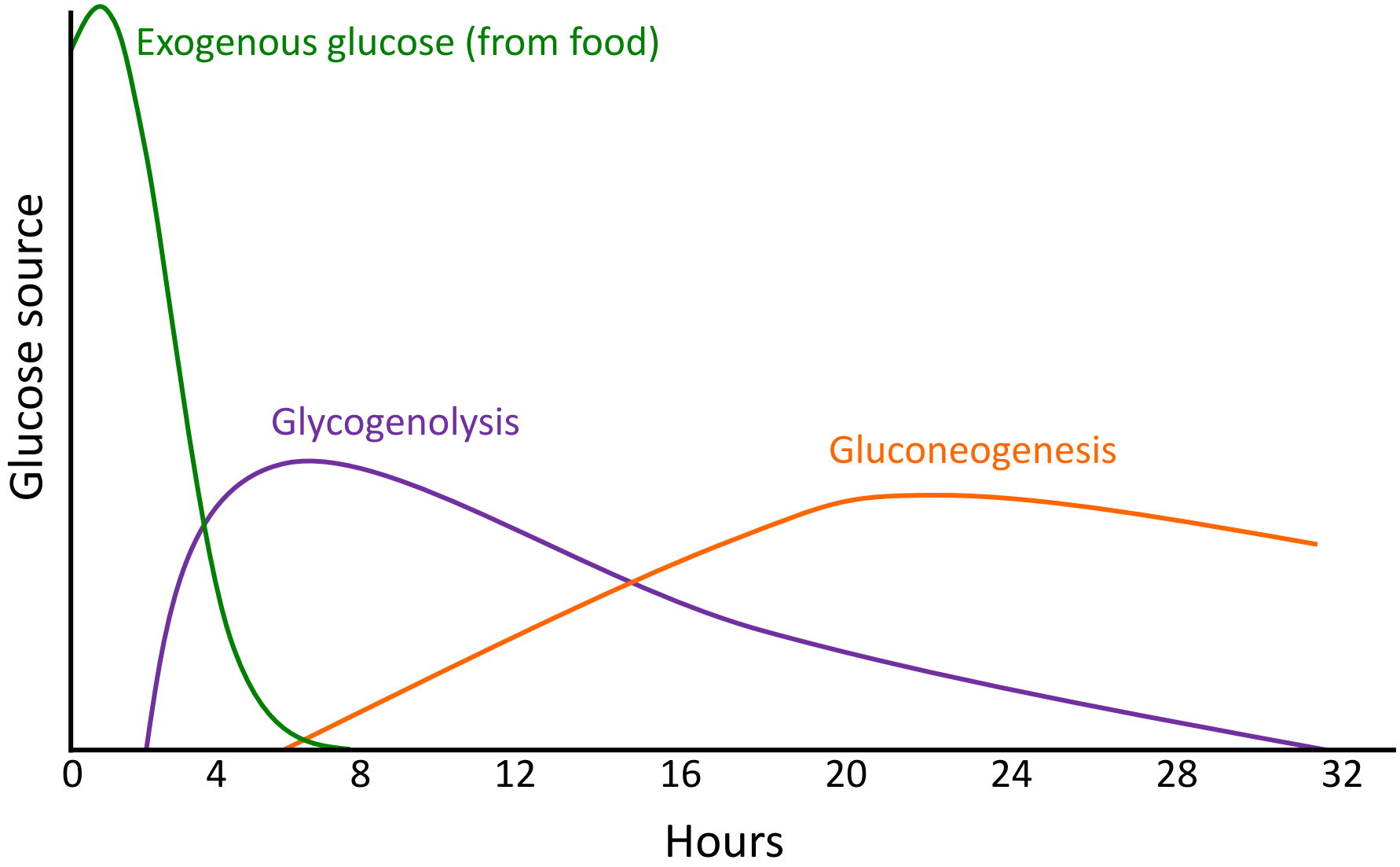
The production of glucose is a necessity

- Glucose is essential for our organism
  - Our body needs 160 g of glucose each day (75% required by the brain)
  - Our body contains 20 g of free glucose and 500 g of glucose stored as glycogen (100 g in the liver and 400 g in muscles)
- The production of glucose is an absolute necessity
- Glycogenolysis and gluconeogenesis allow the generation of glucose during:
  - Fasting
  - Vigorous exercise

Glucose and glycogen / Degradation and synthesis

	Glucose	Glycogen
Degradation	Glycolysis	Glycogenolysis
Synthesis	Gluconeogenesis	Glycogenesis

Glucose sources during fasting



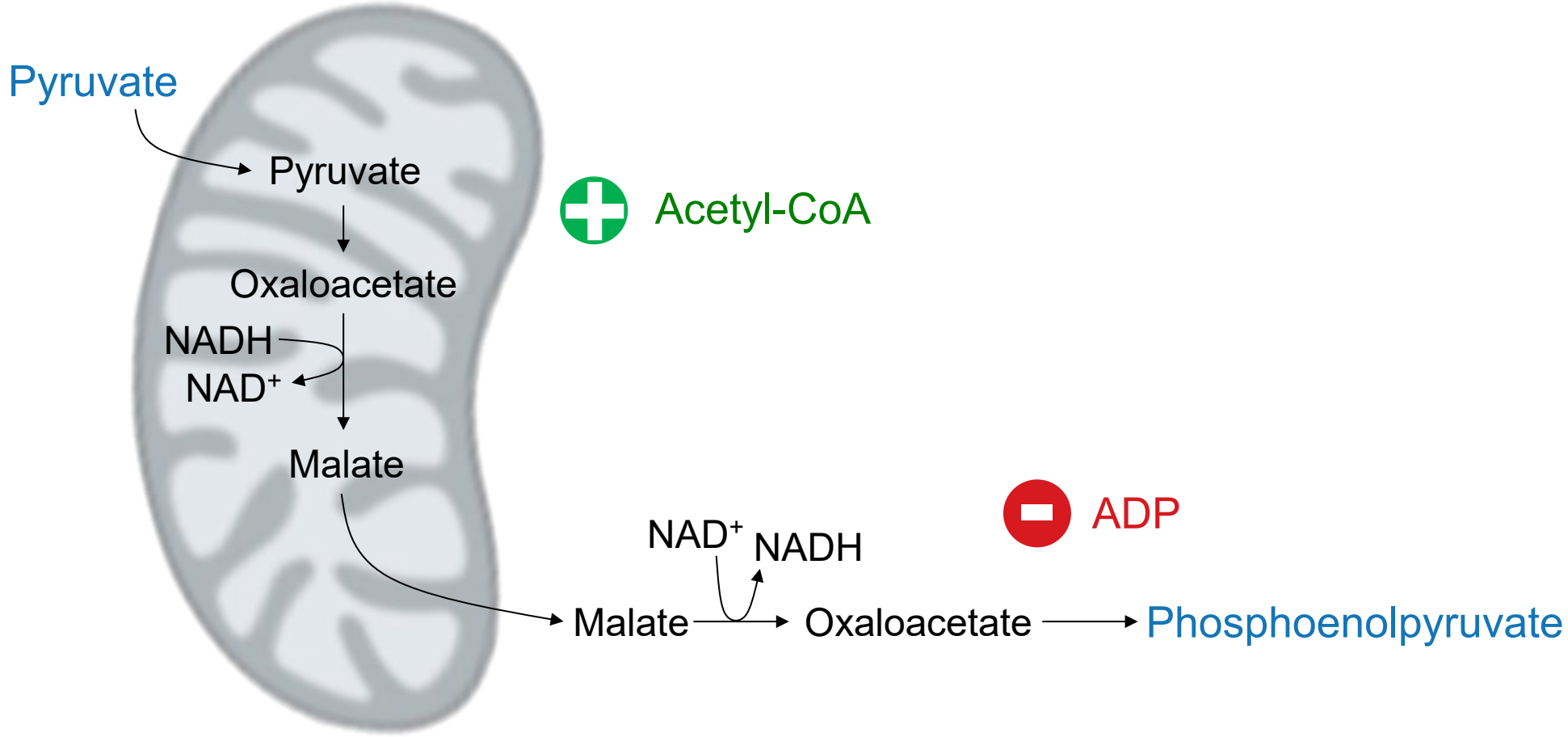
Gluconeogenesis

- Occurs mainly in the liver, and to a less extent in the renal cortex
- In the mitochondria and the cytosol
- Is not the reverse pathway of glycolysis (although the end product of one is the starting compound of the other)
  - In this way, both processes are thermodynamically favorable and have an independent (although coordinated) regulation
  - Seven (reversible) reactions are used in both pathways
  - Irreversible reactions in glycolysis must be bypassed

Bypass steps in gluconeogenesis

- Synthesis of phosphoenolpyruvate from pyruvate
- Formation of fructose 6-phosphate from fructose 1,6-bisphosphate
- Conversion of glucose 6-phosphate to glucose

Synthesis of phosphoenolpyruvate from pyruvate



$$\text{Pyruvate} + \text{ATP} + \text{GTP} + H_2O \rightarrow \text{PEP} + \text{ADP} + \text{GDP} + P_i + 2\,H^+$$

Additional bypasses

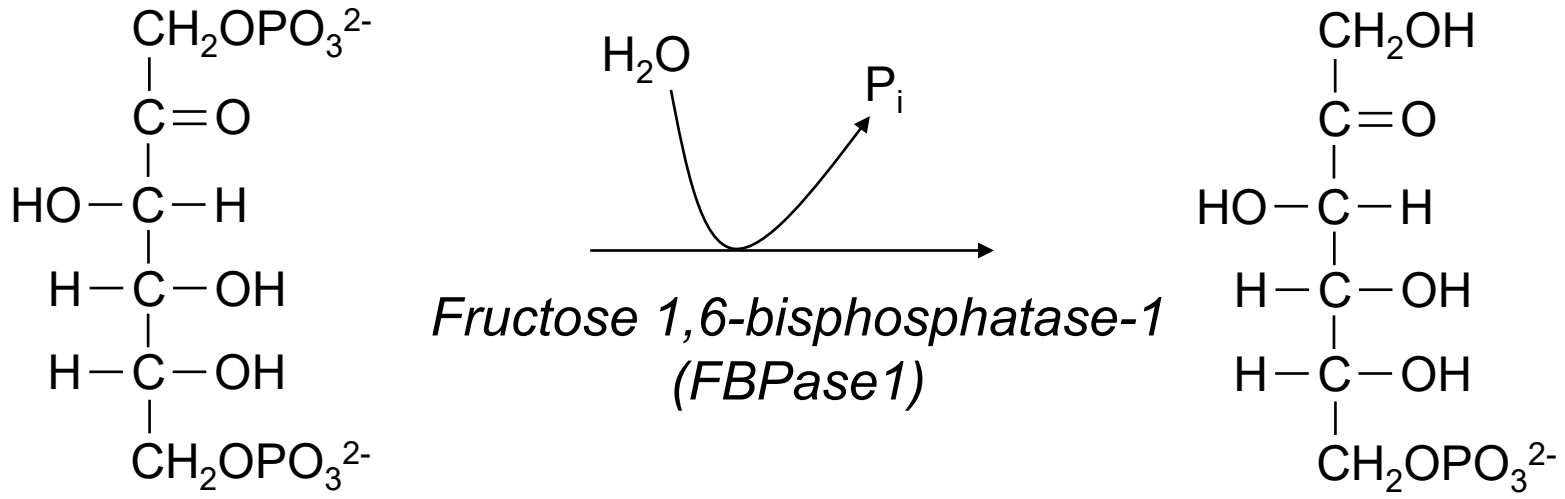
Fructose 1,6-bisphosphate → fructose 6-phosphate

Catalyzed by *fructose 1,6-bisphosphatase-1*

Glucose 6-phosphate → glucose

Catalyzed by *glucose 6-phosphatase*

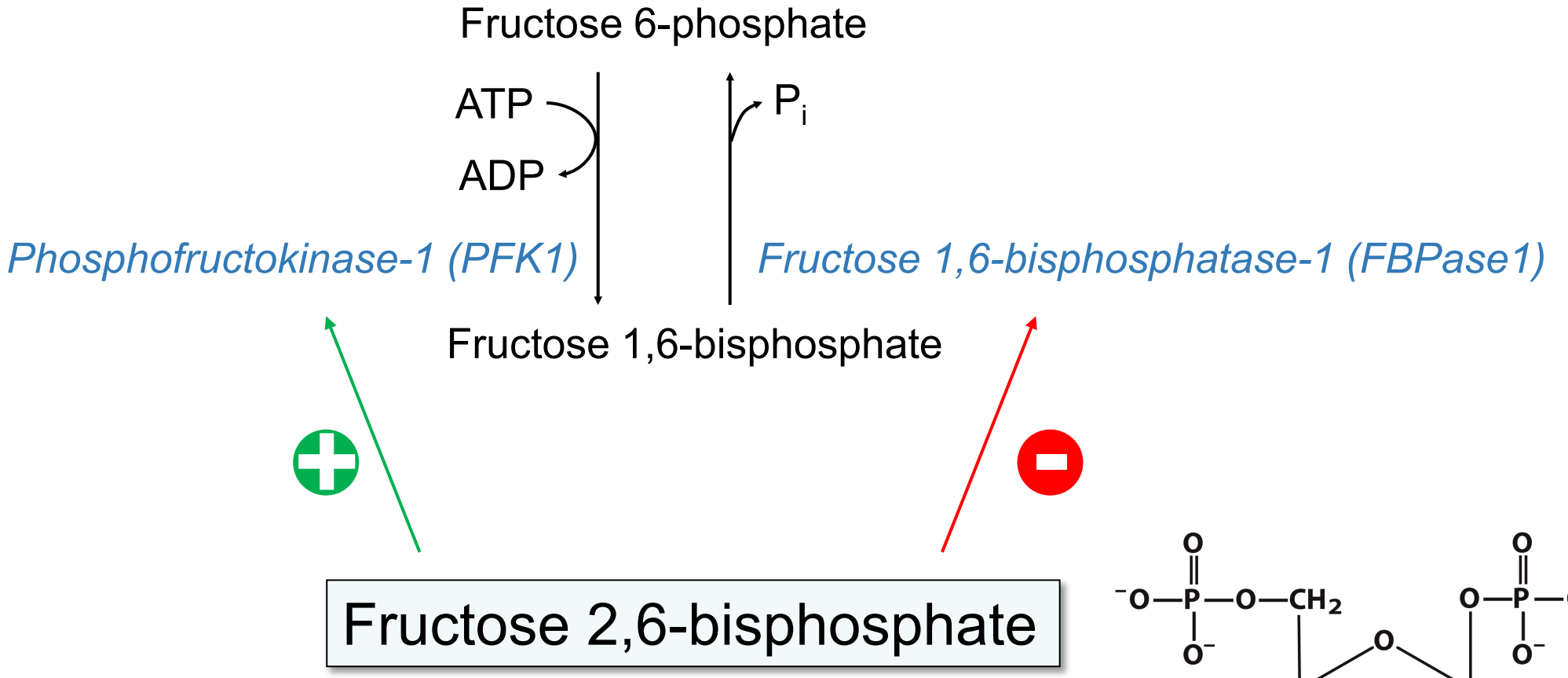
Hydrolysis of fructose 1,6-bisphosphate

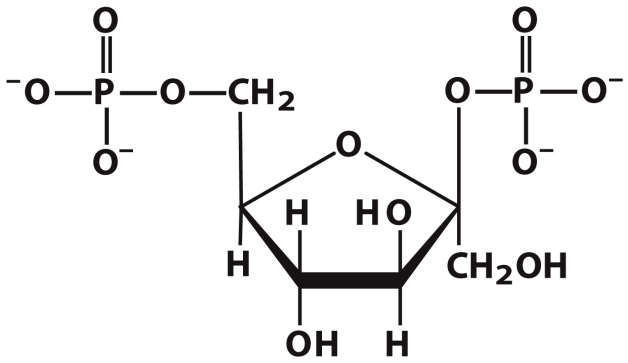


Fructose 1,6-bisphosphate

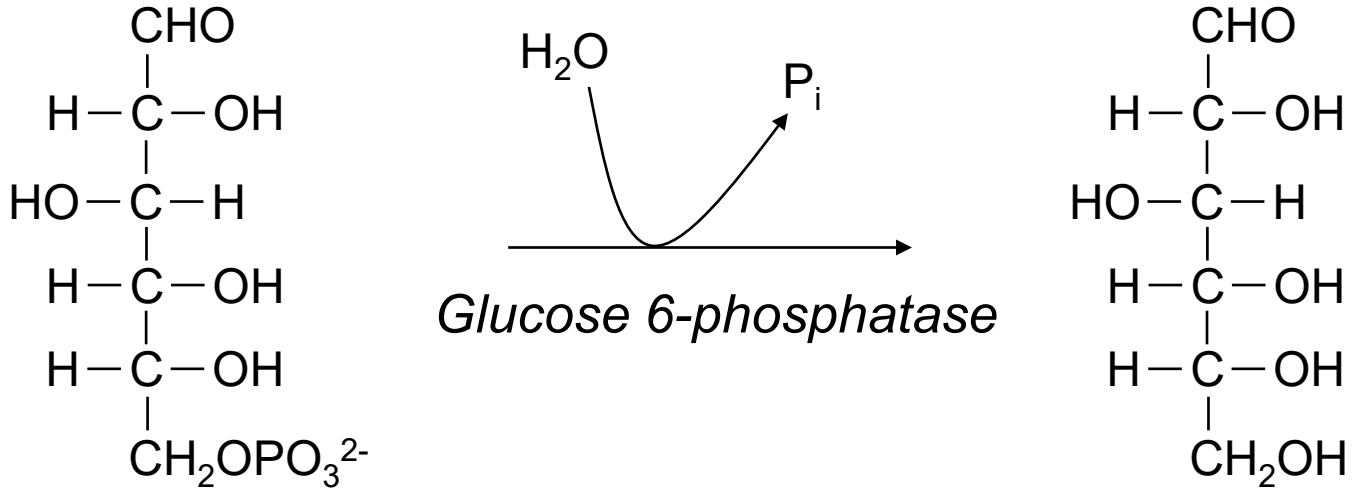
Fructose 6-phosphate

Coordinated regulation of glycolysis and gluconeogenesis





Hydrolysis of glucose 6-phosphate



Glucose 6-phosphate

Glucose

Transport of glucose out of the cell

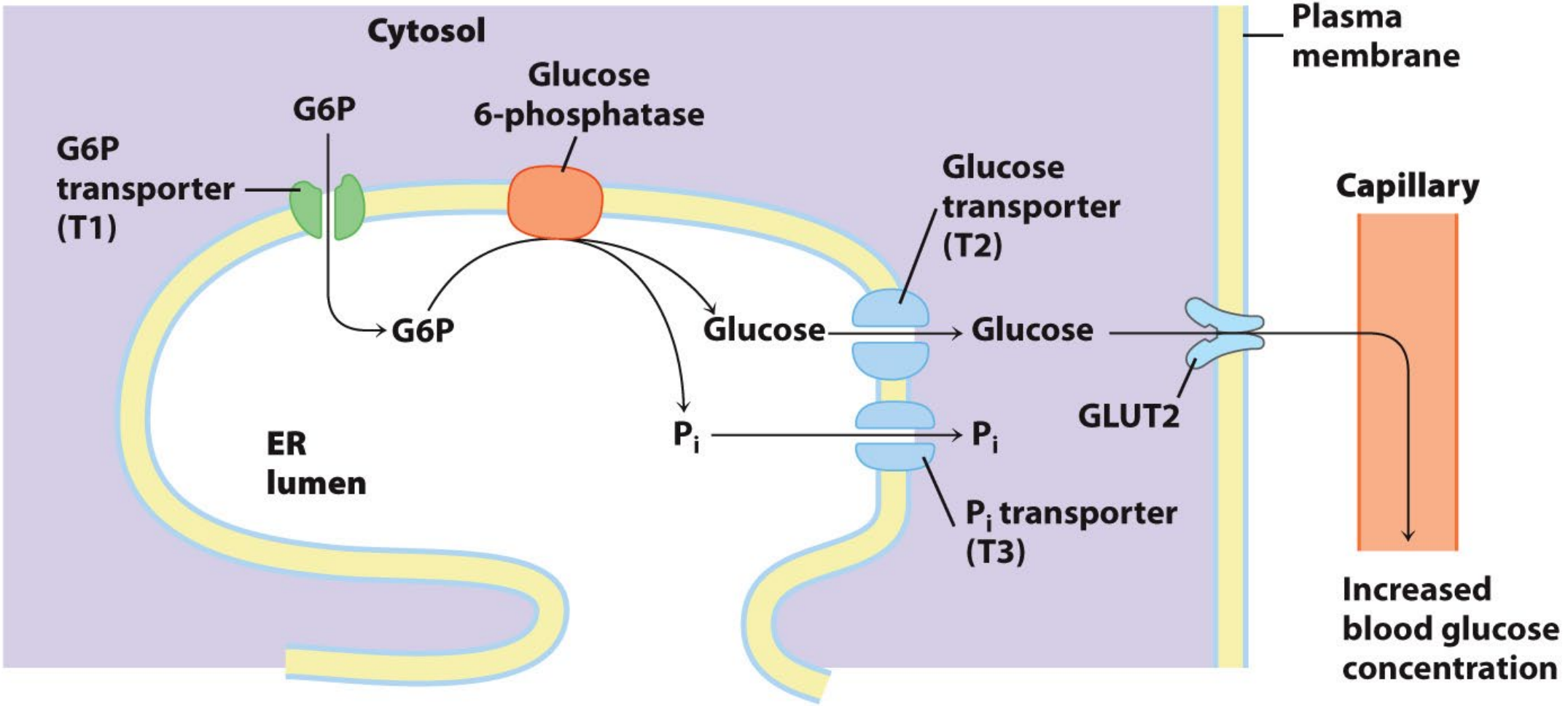


Figure 15-30
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Gluconeogenesis is “expensive”

TABLE 14-3 Sequential Reactions in Gluconeogenesis Starting from Pyruvate	
$\text{Pyruvate} + HCO_3^- + \text{ATP} \longrightarrow \text{oxaloacetate} + \text{ADP} + P_i$	×2
$\text{Oxaloacetate} + \text{GTP} \rightleftharpoons \text{phosphoenolpyruvate} + CO_2 + \text{GDP}$	×2
$\text{Phosphoenolpyruvate} + H_2O \rightleftharpoons \text{2-phosphoglycerate}$	×2
$\text{2-Phosphoglycerate} \rightleftharpoons \text{3-phosphoglycerate}$	×2
$\text{3-Phosphoglycerate} + \text{ATP} \rightleftharpoons \text{1,3-bisphosphoglycerate} + \text{ADP}$	×2
$\text{1,3-Bisphosphoglycerate} + \text{NADH} + H^+ \rightleftharpoons \text{glyceraldehyde 3-phosphate} + NAD^+ + P_i$	×2
$\text{Glyceraldehyde 3-phosphate} \rightleftharpoons \text{dihydroxyacetone phosphate}$	
$\text{Glyceraldehyde 3-phosphate} + \text{dihydroxyacetone phosphate} \rightleftharpoons \text{fructose 1,6-bisphosphate}$	
$\text{Fructose 1,6-bisphosphate} \longrightarrow \text{fructose 6-phosphate} + P_i$	
$\text{Fructose 6-phosphate} \rightleftharpoons \text{glucose 6-phosphate}$	
$\text{Glucose 6-phosphate} + H_2O \longrightarrow \text{glucose} + P_i$	
Sum: $2\ \text{Pyruvate} + 4\text{ATP} + 2\text{GTP} + 2\text{NADH} + 2H^+ + 4H_2O \longrightarrow \text{glucose} + 4\text{ADP} + 2\text{GDP} + 6P_i + 2NAD^+$	

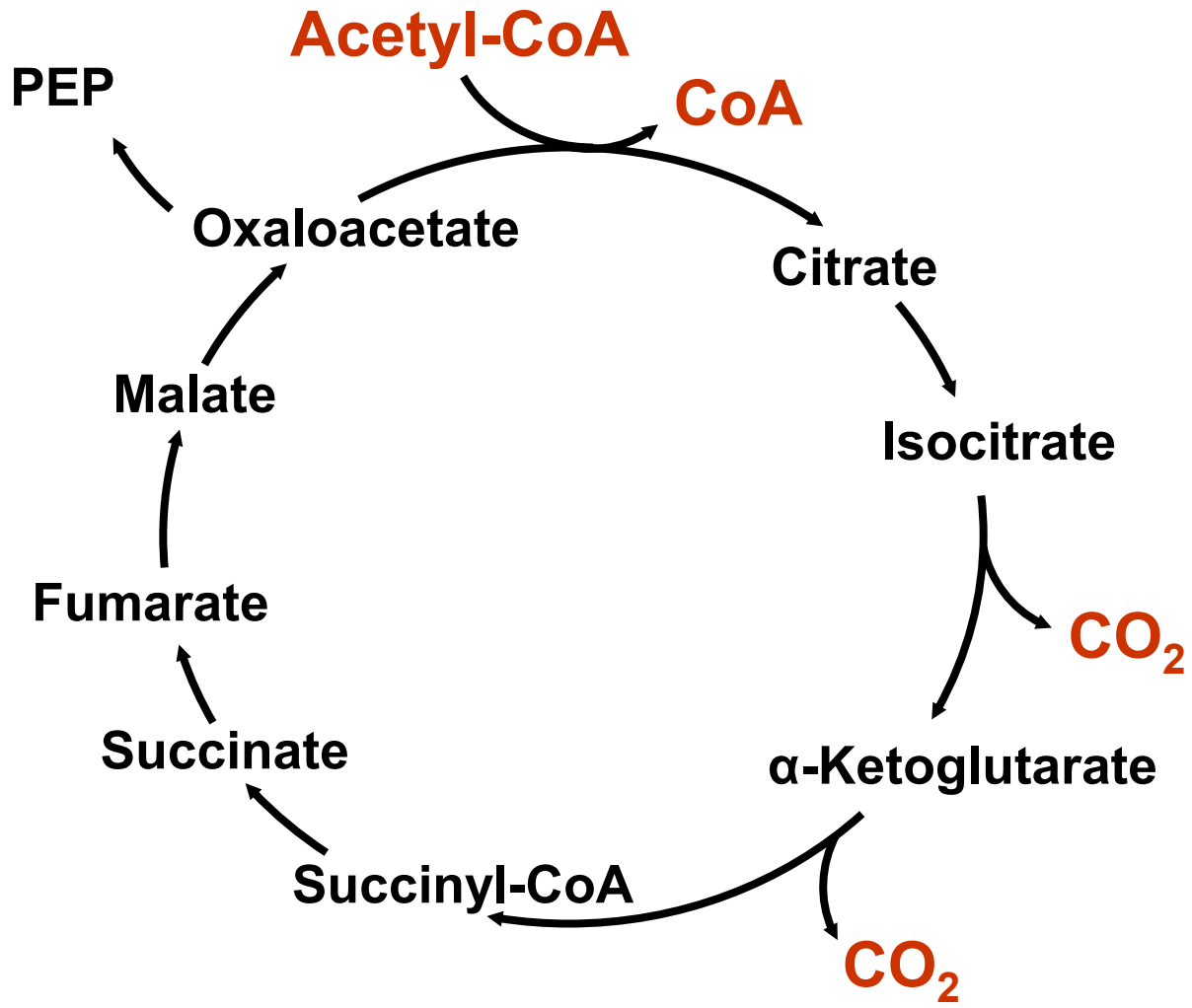
Note: The bypass reactions are in red; all other reactions are reversible steps of glycolysis. The figures at the right indicate that the reaction is to be counted twice, because two three-carbon precursors are required to make a molecule of glucose. The reactions required to replace the cytosolic NADH consumed in the glyceraldehyde 3-phosphate dehydrogenase reaction (the conversion of lactate to pyruvate in the cytosol or the transport of reducing equivalents from mitochondria to the cytosol in the form of malate) are not considered in this summary. Biochemical equations are not necessarily balanced for H and charge (p. 498).

Table 14-3
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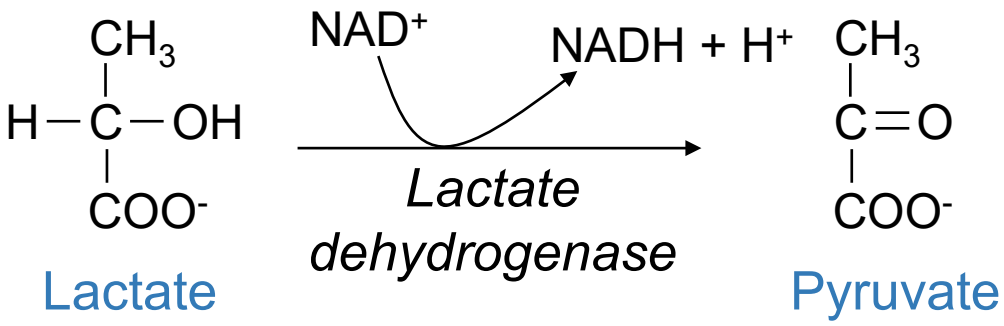
Precursors for gluconeogenesis

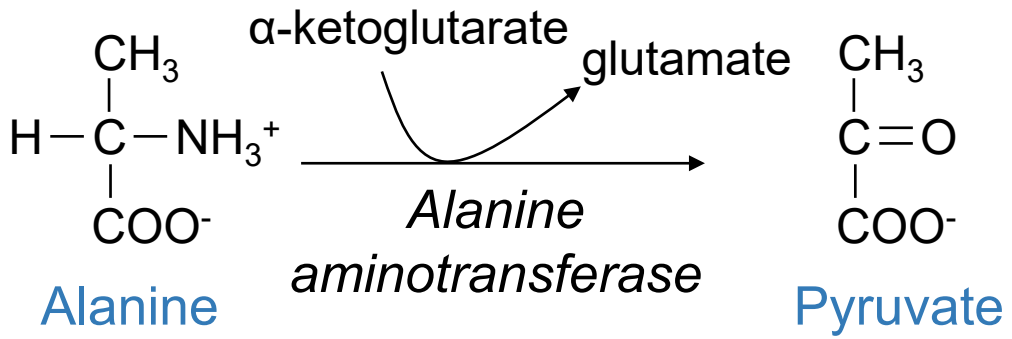
- Animals can produce glucose from glycerol, lactate or most amino acids
- Animals cannot produce glucose from fatty acids

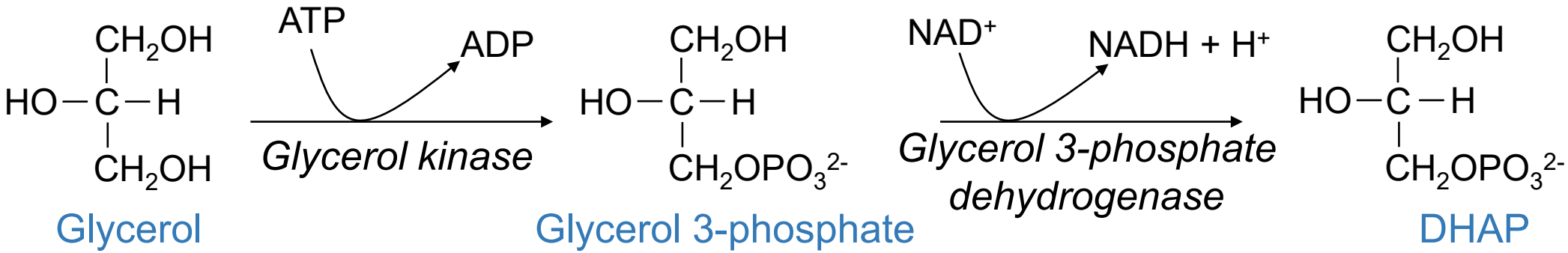
Acetyl-CoA cannot be used for the synthesis of glucose



Precursors for gluconeogenesis







Glucogenic amino acids

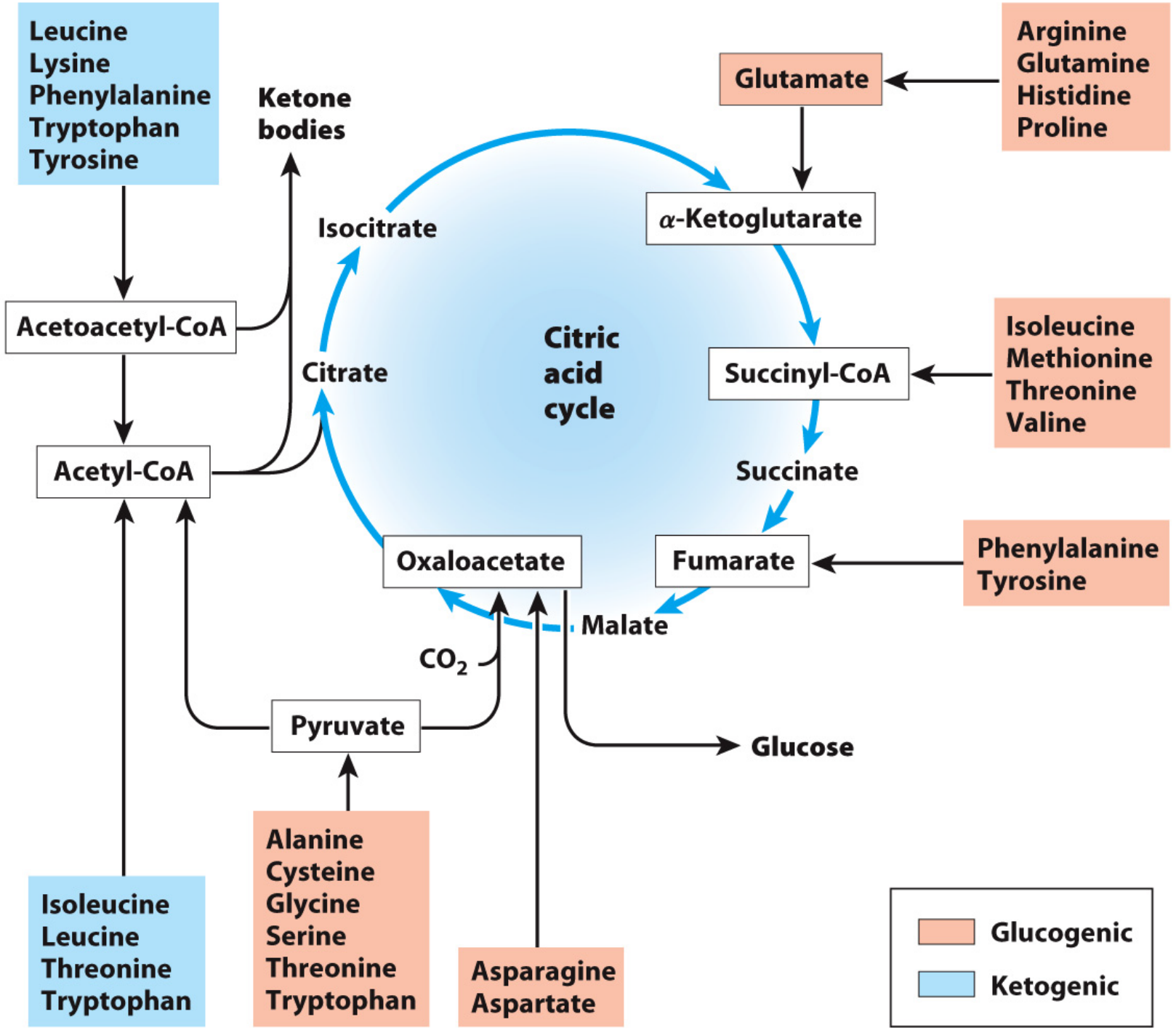
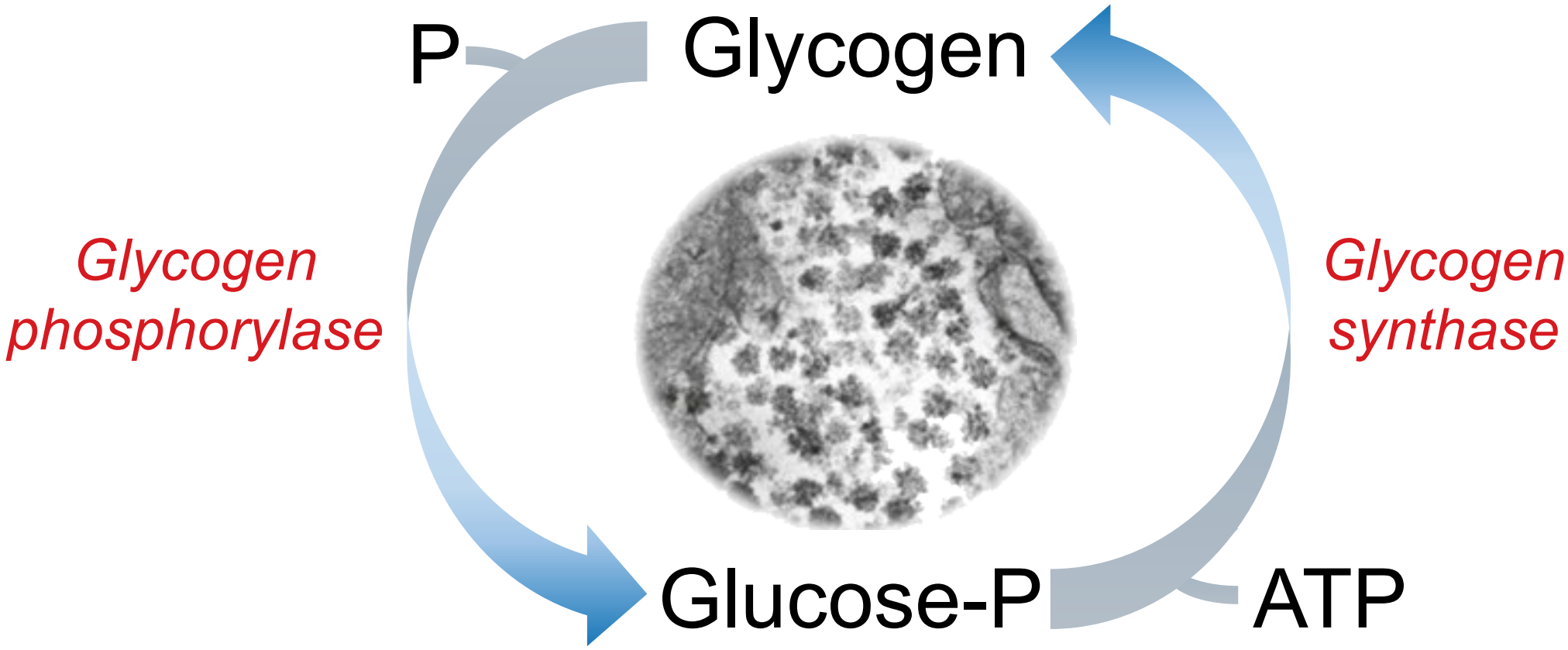


Figure 18-15
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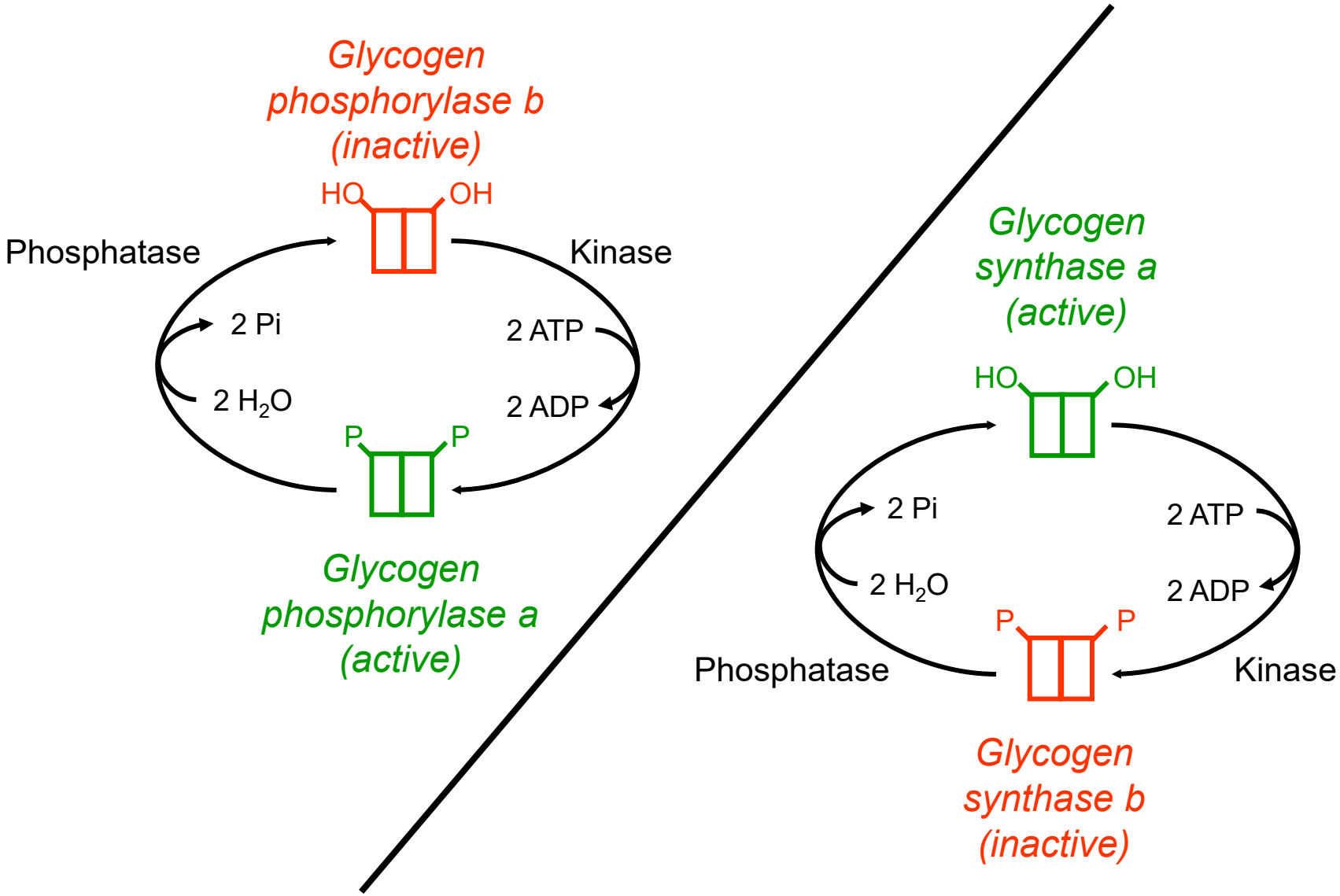
Glycogen metabolism

- Glycogenolysis (glycogen breakdown)
- Glycogenesis (glycogen synthesis)

Glycogen metabolism



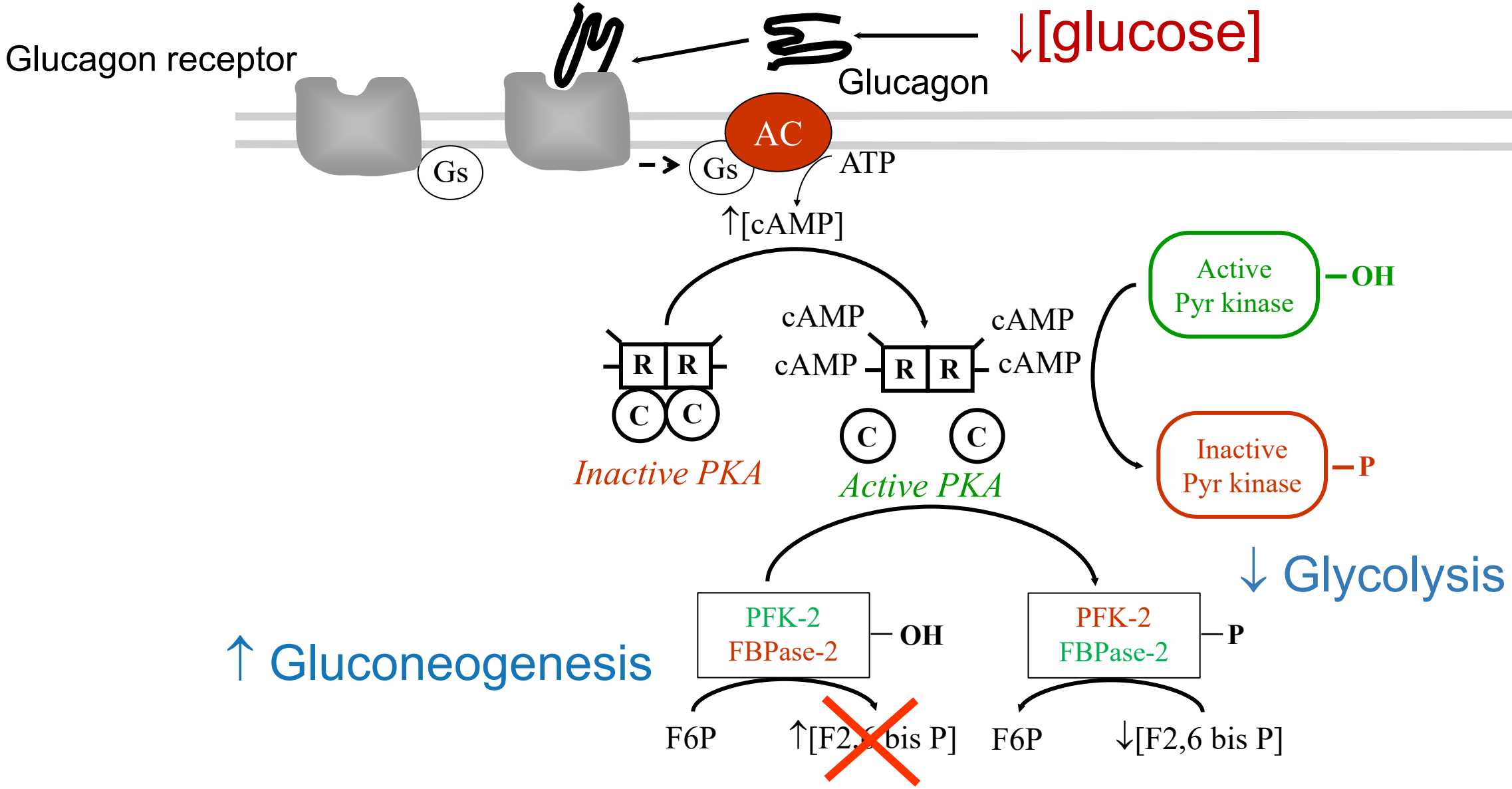
Coordinated regulation of glycogen metabolism



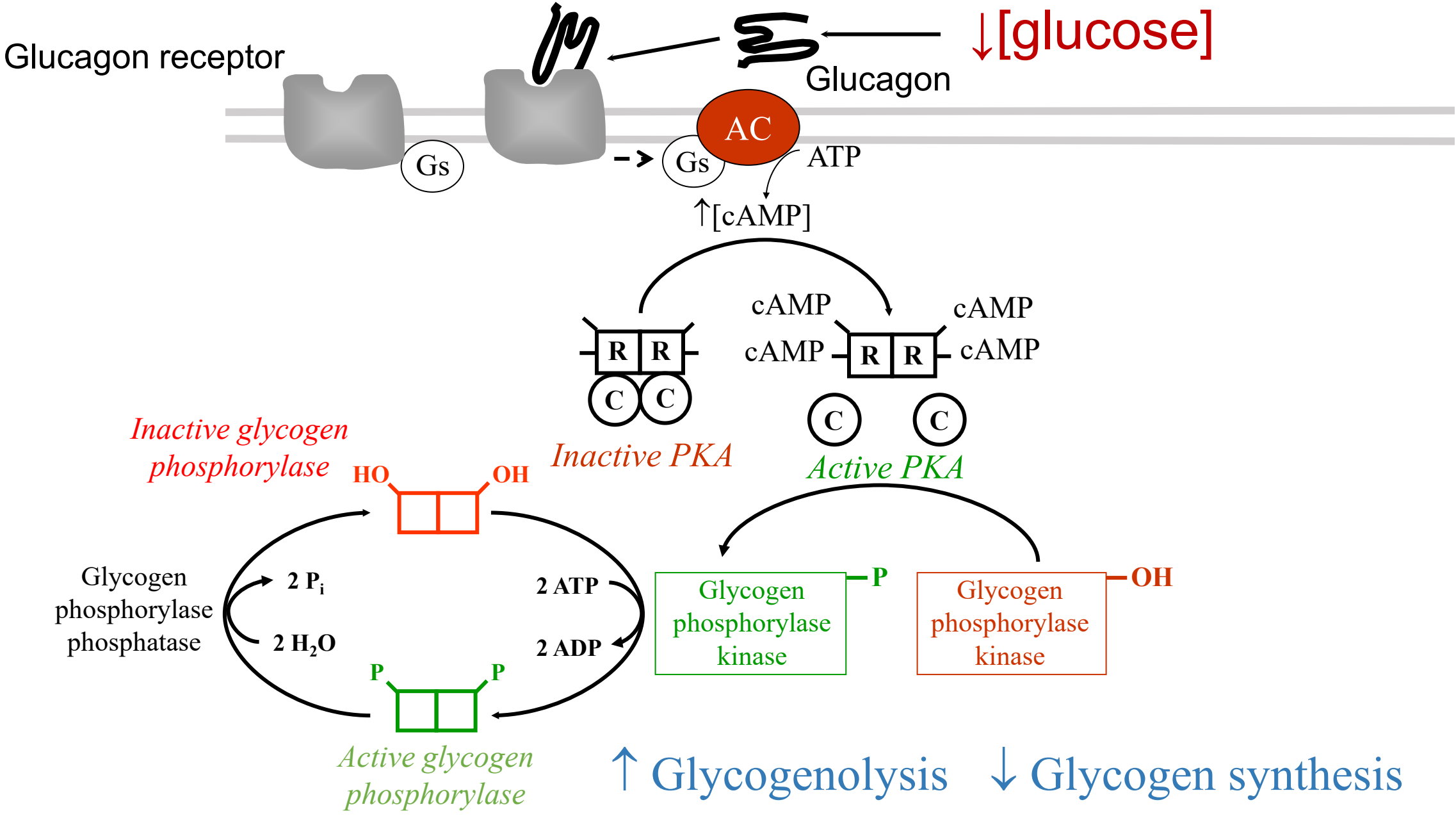
Hormonal regulation of glycogen metabolism

- Epinephrine (myocytes) and glucagon (hepatocytes) stimulate breakdown of glycogen
- Insulin stimulates glycogen synthesis

Liver response to glucose levels



Liver response to glucose levels



Hormonal regulation of glucose metabolism

		Fasting	Carbohydrate meal	Exercise or stress
Liver	Hormone	Glucagon Insulin	Glucagon Insulin	Epinephrine
	Response	Glycogen degradation Glycogen synthesis Glycolysis Gluconeogenesis	Glycogen degradation Glycogen synthesis Glycolysis Gluconeogenesis	Glycogen degradation Glycogen synthesis Glycolysis Gluconeogenesis
Muscle	Hormone	Insulin	Insulin	Epinephrine
	Response	Glycogen degradation Glycogen synthesis Glycolysis	Glycogen degradation Glycogen synthesis Glycolysis Glucose uptake	Glycogen degradation Glycogen synthesis Glycolysis

Team challenge

Glycogen storage diseases

TABLE 1 Glycogen Storage Diseases of Humans

Type (name)	Enzyme affected	Primary organ/cells affected	Symptoms
Type 0	Glycogen synthase	Liver	Low blood glucose, high ketone bodies, early death
Type Ia (von Gierke)	Glucose 6-phosphatase	Liver	Enlarged liver, kidney failure
Type Ib	Microsomal glucose 6-phosphate translocase	Liver	As in type Ia; also high susceptibility to bacterial infections
Type Ic	Microsomal P_i transporter	Liver	As in type Ia
Type II (Pompe)	Lysosomal glucosidase	Skeletal and cardiac muscle	Infantile form: death by age 2; juvenile form: muscle defects (myopathy); adult form: as in muscular dystrophy
Type IIIa (Cori or Forbes)	Debranching enzyme	Liver, skeletal and cardiac muscle	Enlarged liver in infants; myopathy
Type IIIb	Liver debranching enzyme (muscle enzyme normal)	Liver	Enlarged liver in infants
Type IV (Andersen)	Branching enzyme	Liver, skeletal muscle	Enlarged liver and spleen, myoglobin in urine
Type V (McArdle)	Muscle phosphorylase	Skeletal muscle	Exercise-induced cramps and pain; myoglobin in urine
Type VI (Hers)	Liver phosphorylase	Liver	Enlarged liver
Type VII (Tarui)	Muscle PFK-1	Muscle, erythrocytes	As in type V; also hemolytic anemia
Type VIb, VIII, or IX	Phosphorylase kinase	Liver, leukocytes, muscle	Enlarged liver
Type XI (Fanconi-Bickel)	Glucose transporter (GLUT2)	Liver	Failure to thrive, enlarged liver, rickets, kidney dysfunction

Four similar case reports

Case 1. A 16-year-old male presented with cramps and weakness in the limbs

Case 2. A 15-year-old woman presented with intense myalgia in the lower extremities and inability to walk

Case 3. An 11-year-old male with alterations in the morphology of motor unit potentials that could be compatible with myopathy

Case 4. A 5-year-old male, who presented an acute picture of muscle pain in the lower limbs with intense weakness

From López-Martín et al. Anales de Medicina Interna 2001

Case report

A 30-year-old woman with nothing noteworthy in her clinical history. Was referred to the rheumatology department from primary care for the study of persistent creatine kinase (CK) elevation (peak value: 4000 IU/L). She complained of asthenia, fatigue and cramps, and the feeling of a slight loss of strength in her lower limbs following strenuous exercise.

On physical examination, her muscle balance was normal. There was no evidence of myopathy. Laboratory tests showed normal liver function, aldolase and lactate dehydrogenase levels, and autoantibody profile. Given that CK continued to be elevated in follow-up analyses, and based on her report of weakness in her lower limbs following strenuous physical exercise, the decision was made to perform a biopsy of the quadriceps muscle. The results revealed the presence, in the sarcolemma, of multiple clear vacuoles containing eosinophilic granules.

Histochemical study demonstrated the absence of activity of the enzyme *myophosphorylase*, a finding that is compatible with **McArdle disease**.

Treatment was based on the control of physical activity and a diet rich in carbohydrates. The patient was informed of the need to consume foods containing simple carbohydrates 5 min before exercising.

Adapted from Diez-Morrondo C et al. Reumatología clínica 2016

Case report

A seven-month-old female infant, a product of a consanguineous marriage, was admitted for progressive abdominal distention, vomiting, and lethargy for four months. There was no history of seizure or recurrent infections. There was one admission in the past due to similar complaints. Her development was slightly delayed. Her own parents and elder siblings were healthy and alive.

On physical examination, the lethargic infant had a rounded doll's face, fatty cheeks, and a protuberant abdomen; weight below the tenth percentile; and length below the fifth. Systemic examination was unremarkable except for a massive and firm hepatomegaly.

When investigated, complete blood count (CBC) showed low hemoglobin, normal total leukocyte count, and differential leukocyte counts. The urine was negative for reducing substances. Apart from the low blood glucose level, blood chemistry revealed high lactic acid level, hyperlipidemia, hyperuricemia, mild elevation in liver enzymes, normal creatine kinase levels, and arterial blood gases revealed high anion gap metabolic acidosis. A liver biopsy demonstrated a mosaic pattern of enlarged hepatocytes, exhibiting abundant clear cytoplasm containing glycogen and lipid vacuoles.

The patient was diagnosed with **Von Gierke disease** (a deficiency in *glucose 6-phosphatase*). Dietary therapy was started as frequent meals with high carbohydrate content.

Adapted from Raza M et al. Cureus 2017